

tably result in a massive jump in demand for reliable electricity supply. Goldman Sachs Research, part of the global investment banking giant Goldman Sachs, estimates that power demand from data centres will grow 160% by 2030. At present, data centres worldwide account for 1-2% of overall power demand, but that will likely rise to 3-4% by the end of the decade. Thus, there has to be a commensurate increase in power capacity in the country.

But there isn't a lot of time to get these pieces in place if India is to realise its ambition of being a leader in the technology space. Rating agency S&P Global estimates that more than \$100 billion will be invested in data centres in the Asia-Pacific region over the next five years. It sees countries like Malaysia and Vietnam giving India a run for its money.

BIG DATA

Though it lags behind in comparison, India has in fact seen a rapid increase in data centre capacity in the past few years. With the addition of 211 MW of data centre capacity in 2023, total installed capacity stood at 1,011 MW at the end of the year, according to financial services firm Avendus. It sees a \$25-billion market opportunity over the next 10 years and believes that India's data centre capacity will expand at a 26% CAGR to 2,000 MW by December 2026.

Suresh Rathod, President of Colocation at CtrlS Datacenters, says the Indian data centre ecosystem is on the cusp of a great transformation. "The proliferation of data-intensive technologies, such as AI, IoT, and 5G, coupled with increased digital adoption across all sectors, is fuelling an unprecedented surge in demand for data storage and processing capabilities. This growth is no longer confined to major metropolitan areas, as Tier II and III cities are also emerging as data centre hubs, reflecting India's expanding digital economy," he says.

CtrlS currently operates 15 data centres located across eight key markets with a combined capacity of more than 250 MW. "This capacity is being actively expanded, with an additional 200 MW currently under construction and ambitious plans to reach over 1,000 MW by 2029," says Rathod.

Of the 1 GW of installed capacity in India at present, 50% is taken up by hyperscalers or large cloud service providers. Industry insiders expect a big shift in those numbers as other sectors embrace data as a central component of their business. Ashim Sarin, Director & Chief Operating Officer of real estate company Anant Raj Ltd, that has a presence in the data centre segment, says, "The establishment of data centres in India will benefit

ALL THE RAGE

- **Data centre demand has jumped because the average Indian cell phone user consumes over 19 GB data a month, which is the highest globally**
- **There are 245 data centres in India, considerably lesser than the US's tally of 2,987, indicating the growth potential**
- **Government policies like the Digital Personal Data Protection Act, and the growth of e-commerce are driving expansion**
- **Sensing a big opportunity, major conglomerates like the Adani Group and Reliance Industries are investing heavily in this space**
- **Major investment is needed now to improve the energy infrastructure to provide reliable power supply**

